Membership algorithm (CYK) :-

→ The CYK (Cocke - Younger - Kasami) is a algorithm which is based on dynamic programming technique and assumes that the grammar is in ~~CNF~~ CNF.

→ The CYK algorithm is used to check whether the given string x is in the language.

→ The structure of the rules in a chomsky Normal form grammar.

→ Given a CFG, $G = (V_N, V_T, P, S)$ and a string x in $L(G)$

Algorithm :-

Begin

1. For $i = 1$ to n do

$V_{i1} = \{ A / A \to a$ is a production and i^{th} symbol of x is $a \}$

for $j = 2$ to n do

for $i = 1$ to $n - j + 1$ do

begin

$V_{ij} = \phi$

For $k = 1$ to $j-1$ do

$V_{ij} = V_{ij} \cup \{ A \rightarrow BC \mid B$ is in V_{ik},

C is in $V_{i+k, j-k} \}$

end.

end

<u>Procedure:-</u>

→ construct a Triangular Table

→ Each row corresponds to one length of Substring

(i) Bottom row - string of length 1

(ii) Second from Bottom row - String of length 2

(iii) Top row - String w

→ $X_{i,i}$ is the set of variables A such That Such that $A \rightarrow w_i$ is a production of G

→ compare atmost n pairs of previously computed sets:

$(X_{i,i}, X_{i+1,j}), (X_{i,i+1}, X_{i+2,j}) \ldots$

$(X_{i,j-1}, X_{j,j})$

Q:

$X_{1,5}$				
$X_{1,4}$	$X_{2,4}$			
$X_{1,3}$	$X_{2,3}$	$X_{3,3}$		
$X_{1,2}$	$X_{2,2}$	$X_{3,2}$	$X_{4,2}$	
$X_{1,1}$	$X_{2,1}$	$X_{3,1}$	$X_{4,1}$	$X_{5,1}$
w_1	w_2	w_3	w_3	w_5

Table for string 'w' that has length 5.

(1)

$S \to AB \mid BB$

$A \to CC \mid AB \mid a$

$B \to BB \mid CA \mid b$

$C \to BA \mid AA \mid b$

$w = aabb$

Sol. Construct a Triangular Table of string length 4.

$l = 4$	$X_{1,4}$ $\{S, A, B, C\}$			
$l = 3$	$X_{1,3}$ $\{A, C\}$	$X_{2,3}$ $\{S, A, C\}$		
$l = 2$	$X_{1,2}$ $\{C\}$	$X_{2,2}$ $\{S, A\}$	$X_{3,2}$ $\{S, B, A\}$	
length = 1	$X_{1,1}$ $\{A\}$	$X_{2,1}$ $\{A\}$	$X_{3,1}$ $\{B, C\}$	$X_{4,1}$ $\{B, C\}$
	a	a	b	b

i

j

for length 1

→ $X_{1,1} = a = \{A\}$

→ $X_{2,1} = a = \{A\}$

→ $X_{3,1} = b = \{B, C\}$

→ $X_{4,1} = b = \{B, C\}$

for length = 2

→ $X_{1,2} = aa$

$= \{A\}\ \{A\}$

$= \{AA\} \rightarrow C$

→ $X_{2,2} = ab$

$\{A\} \cdot \{B, C\}$

$= \{AB, AC,\}$

$= \{S, A, \phi\}$

$= \{S, A\}$.

→ $X_{3,2} = bb$

$\{B, C\}\ \{B, C\}$

$\{BB, BC, CB, CC\}$

$\{S, B, A\}$

length = 3

$X_{1,3} = aab$

$\{a\} \cdot \{ab\} \cup \{aa\} \cdot \{b\}$

$\{A\} \cdot \{S, A\} \cup \{c\} \cdot \{B, c\}$

$\{AS, AA\} \cup \{cB, cc\}$

$\{\phi, c\} \cup \{\phi, A\}$

$\{A, c\}$

$X_{2,3} = abb$

$= \{a\} \cdot \{bb\} \cup \{ab\} \cdot \{b\}$

$= \{A\} \{S, A, B\} \cup \{S, A\} \cdot \{B, c\}$

$= \{AS, AA, AB\} \cup \{SB, Sc, AB, Ac\}$

$= \{\phi, c, S, A\} \cup \{\phi, \phi, S, A, \phi\}$

$= \{S, A, c\}$

length 4:

$X_{1,4} = aabb$

$= a \cdot abb \cup aa \cdot bb \cup aab \cdot b$

$= \{A\} \cdot \{S, A, c\} \cup \{c\} \cdot \{S, B, A\} \cup \{A, c\} \cdot \{B, c\}$

$= \{AS, AA, Ac\} \cup \{cS, cB, cA\} \cup \{AB, Ac, cB, cc\}$

$\phi \quad c \quad \phi \qquad \phi \quad \phi \quad B \qquad S, A, \phi \quad \phi \quad A$

$\{c\} \cup \{B\} \qquad \{S, A, A\}$

$= \{S, A, B, c\}$

∴ The final set contains a start symbol S as a member. then the string is belongs to the language.

Therefore the the string w = {aabb} is a member of L(G).

(2)

$S \to AB \mid BC$

$A \to BA \mid a$

$B \to CC \mid b$

$C \to AB \mid a$

find the string w is ~~beto~~ member of L(G) or not by using CYK algorithm.

"baaba"

Sol:

L=5	X_{15} C, S, A				
L=4	X_{14} —	X_{24} A, S, C			
L=3	X_{13} —	X_{23} B	X_{33} B		
L=2	X_{12} A, S	X_{22} B	X_{32} B, C	X_{42} A, S	
L=1	X_{11} B	X_{21} A, C	X_{31} A, C	X_{41} B	X_{51} A, C
	b	a	a	b	a